

FELICITY YICK



✉ felicityyick@gmail.com

👤 <https://www.felicityyick.com>

🌐 <https://www.linkedin.com/in/felicity-yick/>

📍 San Francisco Bay Area, CA

ABOUT ME

Creative AR/VR engineer with 5+ years of experience building high-quality, immersive experiences through robust asset pipelines, production-ready tooling, and artist-centric workflows. Proven track record working with Unreal Engine, AI-driven creative tools, scripting, and real-time interactive systems, bridging engineering and art to ship scalable, performance-critical products.

EXPERIENCE

Meta Reality Labs

San Francisco Bay Area, CA (Hybrid)

Technical Artist (Contract through Magnit)

Jan 2025 - Present

- On the Horizon OS Art Team, I play a key role in shipping the switchable **Immersive Home (v85, Connect 2025)** by partnering cross-functionally with Product, Art, and Engineering to deliver high-quality, performance-critical VR content on Quest 2 and Quest 3.
- Owned **art ingestion, optimization, and shader development**, ensuring the five immersive environments met high visual quality bars while remaining within strict performance and memory constraints for standalone headsets.
- Expanded, hardened, and productionized our internal **Unreal Engine 5.1 / 5.5 Python and C++ plugins** to prepare assets for Meta's internal engines, including lightmap baking, USD export with LODs, packed texture workflows, and PBR/unlit material baking; enabled artists and external vendors to ship content reliably with minimal friction.
- Drive R&D investigations into **Gaussian Splatting**, prototyping rigid body animated splat demos to evaluate feasibility for Immersive Home and future XR experiences

TikTok

San Francisco Bay Area, CA (In Person)

Mid-Level AR Interactive Engineer

Aug 2022 - Dec 2024

- Core engineer on TikTok Effect House's **"Create with AI"**, a platform feature enabling AI-assisted generation of AR effects; supported **10K+ effects**, primarily created by first-time users, materially reducing creator onboarding friction.
- Designed and built a **scalable, AI-driven system** for parameterized and gamified AR effects, implementing a modular TypeScript framework that auto-generates visual scripting logic to drive gameplay flow at scale.
- Owned feature development from concept to production, partnering with **Product, Art, and Engineering** to define requirements, execute delivery, and establish best practices in a rapidly evolving cross-functional pipeline.
- Shipped 50+ high-impact AR effects** for organic and branded campaigns, reaching **10M+ users globally**, while consistently optimizing for strict mobile performance and device constraints.
- Represented the Effect House team externally, serving as a lead speaker at **TikTok's first Open House**, presenting at conferences, and providing hands-on technical guidance and support to the global Effect House creator community.

University of Pennsylvania

Philadelphia, PA

Computer Graphics Research Intern

May 2019 - Aug 2020

- Contributed to a research project **simulating crowd processions** in ancient civilizations, developing a parameterized **3D/AR visualization system** for interactive exploration.
- Reconstructed a historically accurate 3D environment of Pachacamac, Peru, authoring **MEL/Python tooling** to convert SketchUp drawings into production-ready assets in Autodesk Maya and Unreal Engine 4.
- Built **interactive controls in UE4 (Blueprints)** enabling real-time manipulation of time-of-day, weather, and material decay, supporting rapid experimentation and visual analysis.
- Modeled, textured, rigged, and animated** period-accurate clothing and characters, implementing idle and locomotion cycles to drive agent-based movement within the simulation.

PROJECTS

Procedural Terrariums

Tools: Houdini

- Using Houdini, designed and built a **procedural generation system** for terrariums, succulent plants, and décor, enabling highly customizable designs through seeded randomization and **artist-directed controls**.
- Implemented **painted-mask workflows** allowing users to explicitly define spawn regions for plant types and terrarium features, balancing procedural scalability with creative intent.

Mini Minecraft

Tools: C++, QtCreator

- Co-developed a 0 to 1 stylized replica of Minecraft, owning **procedural terrain generation** using noise functions, custom biome and texture systems, block animations, and efficient GPU rendering across multiple vertex buffer objects.

EDUCATION

University of Pennsylvania

B.S.E in Computer Graphics | GPA: 3.84

- Interdisciplinary major in computer science and visual arts

Involved in: Co-President of ACM SIGGRAPH chapter, Teaching Assistant (Computer Graphics, 3D Modelling)

TECHNICAL SKILLS

JavaScript, TypeScript, GLSL, Lua, Jest, C++, React, HTML/CSS, Java, Python, Agile development, Git, Perforce, Prompt Engineering, Claude

CREATIVE SKILLS

Autodesk Maya, ZBrush, Substance Painter, Adobe, Creative Suite, Unreal Engine ComfyUI, Houdini